SwipeNest AI — Product Requirements Document (PRD)

Concept: Tinder meets AI-powered apartment hunting.
Market: Paris | **Platform:** Web application | **Stage:** MVP / POC

Part 1 — Functional Requirements

1. Product Vision

SwipeNest is an AI-powered apartment hunting platform that learns users' preferences through swipes, recommends increasingly relevant properties, detects suspicious listings, and simplifies rental applications.

Goal: Make apartment hunting faster, smarter and safer.

2. Core Features

1. Personalized Onboarding

- Users define their budget, preferred neighborhoods, minimum surface and apartment type.
- They can describe their dream apartment in natural language.

2. Tinder-Style Apartment Discovery

- Apartment cards with photos, price, location, surface and description.
- Swipe right to like, left to reject.
- Save favorites and revisit previous listings.
- After a negative swipe, users can optionally select a reason: too expensive, too small, too dark, disliked neighborhood or other.

3. AI-Powered Recommendations

- AI learns from likes, dislikes, rejection reasons and visual preferences.
- Recommendations continuously adapt to user behavior.
- Each apartment receives a personalized match score.
- A **"Why this apartment?"** button explains why the property was recommended and which characteristics best match the user's preferences.

4. AI Taste DNA

- Dashboard showing preferences discovered by AI: natural light, interior style, space, architecture, etc.
- Preferences evolve dynamically after each swipe.
- Users can adjust their preferences manually.

5. Dream Apartment Search

- Search using natural language, e.g. *"Bright Haussmannian apartment with wooden floors under €1,500."*
- AI understands the request and ranks the most relevant listings.

6. AI Scam Detective

- Identifies suspicious prices, unusual requests and warning signs.
- Displays a risk level (Low / Medium / High) with explanations.

7. AI Application Copilot

- Generates personalized landlord messages using the user profile and apartment details.
- Prepares a rental-document checklist.
- Users review, edit and copy their applications before sending.

8. Additional Features

- Interactive Paris map with apartment locations.
- AI comparison of shortlisted apartments.
- Budget compatibility indicators.

3. User Journey

Create profile → Discover apartments → Swipe → Optionally explain rejection → AI learns → Explore better matches → Understand recommendations → Check risks → Save favorites → Prepare application.

4. Design Direction

Modern, premium, mobile-first interface inspired by Tinder, Airbnb and contemporary AI products.

Smooth swipe animations, beautiful apartment photography, intuitive navigation and visually engaging AI insights.

Part 2 — Technical Specifications

5. Technology Stack

Component	Technology
Frontend	Next.js, React, TypeScript
UI	Tailwind CSS, shadcn/ui, Motion
Backend	Python, FastAPI
Database	SQLite
AI text embeddings	Sentence Transformers
AI image understanding	CLIP (Hugging Face)
ML / ranking	NumPy, scikit-learn
Generative AI	Configurable LLM API
Maps	Leaflet + OpenStreetMap

6. AI Architecture

A. Adaptive Recommendation Engine

- Generate text embeddings from descriptions using Sentence Transformers.
- Generate image embeddings using CLIP.
- Combine text, visual and structured attributes into a personalized ranking.
- Update the user preference vector after each positive or negative swipe.
- Use explicit rejection reasons to adjust relevant preference weights and constraints.
- Dynamically rerank unseen listings based on learned preferences.

B. Explainable AI Recommendations

- Calculate and store contributing factors for each recommendation.
- Identify the strongest matches: budget, location, size, architectural style and learned preferences.
- Power the **"Why this apartment?"** button with an explanation based on actual ranking factors.
- Highlight both matching characteristics and meaningful trade-offs.

C. AI Taste DNA

- Extract interpretable property characteristics from descriptions and images.
- Update preference indicators based on user behavior and rejection reasons.
- Display learned preferences and changes over time.

D. Natural Language Search

- Convert user queries into semantic representations.
- Extract hard constraints (budget, location, surface).
- Rank matching listings using semantic and visual similarity.

E. Scam Detection

- Compare rental prices against local reference data.
- Detect suspicious language and inconsistencies.
- Produce an explainable risk score using weighted indicators.

F. Application Generator

- Use an LLM to generate contextual rental application messages.
- Ground messages in user-provided and listing information.
- Never invent personal information or automatically send applications.

7. Dataset

Primary source: ListingGenie — Hugging Face

- Use 300–500 synthetic listings with images.
- Adapt locations, rental prices and attributes to Paris.
- Include different property styles, neighborhoods and budgets.
- Add intentionally suspicious listings for fraud-detection testing.
- Document synthetic data transformations and preserve attribution.

Precompute AI embeddings for fast recommendations.

8. Technical Requirements

- Fully functional and responsive application.
- Real AI-driven recommendations, not predefined swipe sequences.
- Persistent user preferences, swipe history and optional rejection reasons.
- Fast ranking updates after each interaction.
- Recommendation explanations must reflect actual scoring factors, not fabricated AI justifications.
- Reproducible dataset and preprocessing pipeline.
- Clear separation between frontend, backend and AI modules.
- Secure API-key handling and no real personal documents required.
- README with installation instructions, dependencies and AI tools used.

9. Success Criteria

- Users can swipe through at least 100 unique listings.
- AI recommendations visibly adapt to user preferences.

- Rejection reasons influence subsequent recommendations.
- Users can understand why an apartment was recommended.
- Visual and textual preferences influence matching.
- Natural-language search returns relevant apartments.
- Suspicious listings trigger explainable warnings.
- AI generates editable, personalized applications.
- AI ranking is evaluated against a non-personalized baseline.
- The application can be run locally using the provided code and data.

Product ambition: Build an intelligent apartment discovery experience where AI learns what users truly want, explains its recommendations, protects them from suspicious listings and simplifies the journey from discovery to application.